

11/19/25



PRIORITY

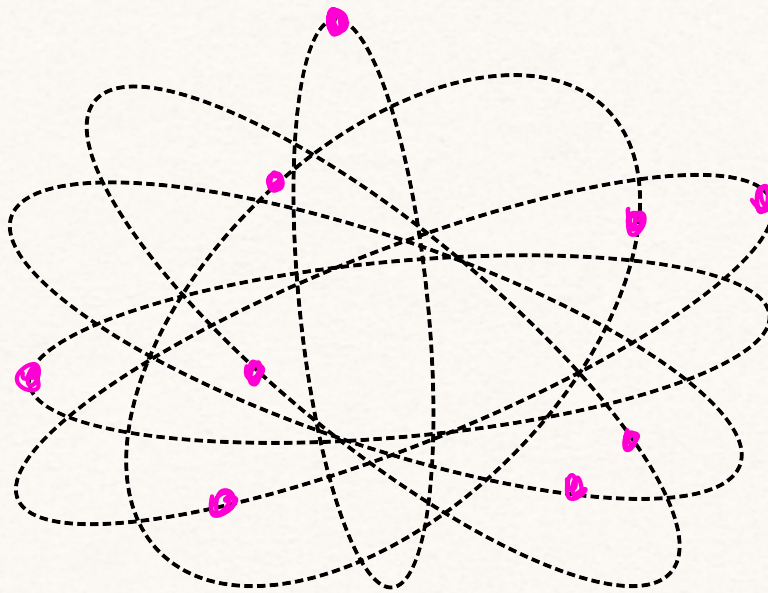


Galaxies Midterm
Corrections

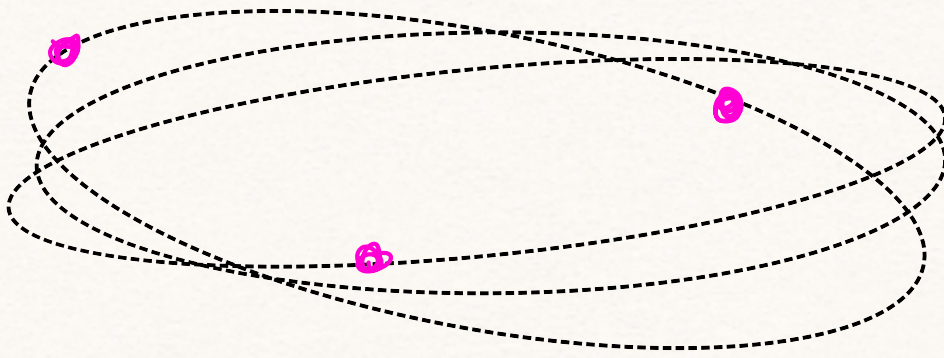
Part 1: Dynamics from Velocity Measurements

Exercise 1: The Faber-Jackson Relationship

a. Velocity dispersion is when you take the average velocities of all the objects you are looking at in a system and understand the statistical distribution of those average velocities. In a galaxy, a high velocity dispersion relates to the spread of the velocities of the stars and would look like this:



A galaxy with low velocity dispersion would have a graph that instead looks like this:



The morphology of a galaxy with low velocity dispersion would be in the shape of a spiral galaxy whereas high velocity dispersion galaxies take on the shape of an elliptical galaxy.

b. L^* represents the total luminosity of the object we are observing. An approximate power law relationship between σ and L^* is:

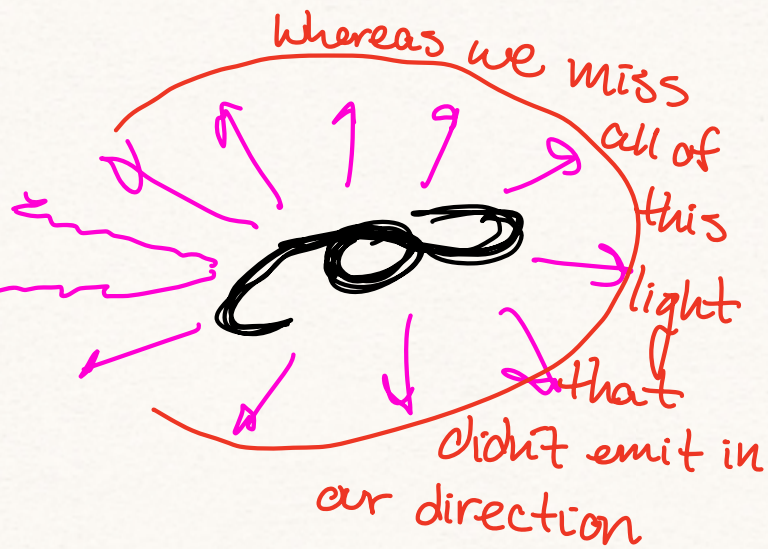
$$\log(L^*) \propto -\log(\sigma) + \log\left(\frac{2.1}{2.2}\right)$$

c. The total flux we observe is only a small fraction of the actual flux that was produced

by the galaxy:



We only see this
line of sight flux



We can use

spectroscopy to measure the velocity dispersion of a galaxy. If we know both the flux and velocity dispersion, we can determine the distance the galaxy is from us.

$$d. \quad \rho = \frac{M}{V} = \frac{M}{\frac{4}{3}\pi R^3} = \frac{3M}{4\pi R^3} \quad \left[\begin{array}{l} \text{Density of sphere with} \\ \text{mass } M \text{ and radius } R \end{array} \right]$$

$$dV = 4\pi r^2 dr$$

$$U = \frac{1}{2} \int_{\mathcal{R}} \rho(r) \Phi(r) d^3r = \frac{1}{2} \int \rho(r) \left(-\frac{3GM}{2R} + \frac{GM}{2R^3} r^2 \right) dV$$

$$U = \frac{1}{2} \int_0^R \rho(r) 4\pi r^2 dr \left(-\frac{3GM}{2R} + \frac{GM}{2R^3} r^2 \right)$$

$$= 2\pi \rho(r) \int_0^R \left(-\frac{3GM}{2R} r^2 + \frac{GM}{2R^3} r^4 \right) dr$$

$$= 2\pi \rho(r) \left[-\frac{3GM}{2 \cdot 2R} r^3 + \frac{GM}{2 \cdot 5 R^3} r^5 \right] \Big|_0^R$$

$$= 2\pi \rho(r) \left[-\frac{GM R^2}{2} + \frac{GM R^2}{10} \right] = 2\pi \rho(r) \left[-\frac{5GM R^2}{10} + \frac{GM R^2}{10} \right]$$

$$= \cancel{2\pi} \left(\frac{3M}{\cancel{4\pi R^3}} \right) \left(\frac{-4GM R^2}{\cancel{10}} \right) = -\frac{3GM^2}{5R} = U$$

e. $KE = \frac{1}{2} m \vec{v}^2$ for a single particle.

$$K = \frac{1}{2} \sum_{i=1}^N m_i \vec{v}_i^2 \text{ for } N \text{ amount of particles}$$

f. Because we are assuming the galaxy is isotropic, we can also assume that the velocity dispersion in the x and z direction is the same.

The total velocity dispersion would then be:

$$\sigma_{\text{tot}}^2 = \sigma_x^2 + \sigma_y^2 + \sigma_z^2$$

$$\sigma_{3D} = \langle v_{xyz} \rangle = \sqrt{\sigma_x^2 + \sigma_y^2 + \sigma_z^2} = \sqrt{3\sigma^2}$$

$$K_{\text{tot}} = \frac{1}{2} M (\langle v_{xyz} \rangle)^2 = \frac{1}{2} M (\sqrt{3\sigma^2})^2$$

$$K_{\text{tot}} = \frac{3}{2} M \sigma^2$$

g. We can use the virial theorem to relate K and U . The relationship between M , R , and σ^2 by:

$$2K + U = 0$$

$$2\left(\frac{3}{2} M \sigma^2\right) - \frac{3GM^2}{5R} = 0$$

$$3M\sigma^2 - \frac{3}{5} \frac{GM^2}{R} = 0$$

$$3M\sigma^2 = \frac{3}{5} \frac{GM^2}{R}$$

$$\sqrt{\sigma^2} = \sqrt{\frac{GM}{5R}}$$

$$\sigma = \pm \sqrt{\frac{GM}{5R}}$$

Velocity dispersion is affected by the mass and radius of a galaxy. Larger M gives us a larger velocity dispersion and a larger radius will give us lower velocity dispersion.

h. Because $M = \kappa L$, the relationship between R , L , and σ^2 is

$$\cancel{GM}\sigma^2 = \cancel{\frac{3}{5}} \frac{GM^2}{R} \Rightarrow \sigma^2 = \frac{GM}{5R}$$

$$M = \frac{5R\sigma^2}{G} \Rightarrow \kappa L = \frac{5R\sigma^2}{G}$$

$$L = \frac{5R\sigma^2}{\kappa G} \Rightarrow L \propto R\sigma^2$$

As radius increases, so does luminosity. As velocity dispersion increases, so does luminosity.

i. $B = \frac{L}{4\pi R^2} \Rightarrow \langle B \rangle = \frac{\langle L \rangle}{4\pi R^2}$ ← Average surface brightness

Now getting rid of R :

$$\sqrt{R^2} = \sqrt{\frac{L}{4\pi B}} \Rightarrow R = \pm \sqrt{\frac{L}{4\pi B}}$$

$$L = \frac{5\left(\sqrt{\frac{L}{4\pi B}}\right)\sigma^2}{\kappa G}$$

$$L^2 = \frac{5^2 \left(\frac{L}{4\pi B} \right) \sigma^4}{\kappa^2 G^2}$$

$$L^2 = \frac{25 L \sigma^4}{4\pi B \kappa^2 G^2}$$

$$L = \frac{25}{4\pi B \kappa^2 G^2} \sigma^4$$

This satisfies the
 $L \propto \sigma^4$ from earlier!

This shows us that our relationship we found in part b is very similar!

j. Throughout this problem, we have assumed that the galaxies that we are looking at are isotropic, however, the scatter that we see in the graph can be due to the anisotropies in the elliptical galaxies. A correction that can be made to the Faber Jackson relationship is adding in effective radii, and surface brightnesses to add realism to the relationship.

Exercise 2: Constrained Cosmological Simulations

a. The subplot for $a = 0.12$ displays the earliest cosmic time because it shows how "homogenous" the universe was, since the far right subplot, where $a = 1.0$, shows the cosmic web which formed later. " a " represents the cosmological expansion factor. The redshift for each panel:

$$a = 0.12$$

$$z = \frac{1}{0.12} - 1 \approx 7.3$$

$$a = \frac{1}{1+z}$$

$$\frac{1}{a} = 1+z$$

$$z = \frac{1}{a} - 1$$

$$a = 0.30$$

$$z = \frac{1}{0.30} - 1 \approx 2.3$$

$$a = 1.00$$

$$z = \frac{1}{1} - 1 = 0$$

b. At higher redshifts, the Universe's density

was more "smooth" than what we see at lower redshifts. Earlier in the Universe, the densities were more homogeneous however, some anisotropies existed. These anisotropies allowed for gravity to grow those over-dense regions, creating halos that could then turn to nodes, filaments, and walls.

As these regions became more dense, they also left voids in their wake. These structures then turned into the cosmic web that we see today, which you can see in the rightmost panel.

C. Comoving coordinates do not include the expansion factor when calculating distances whereas physical coordinates do. The relationship between the two is as follows:

$$\vec{r}(t) = a(t) \vec{x}(t)$$

$$\frac{d}{dt} \vec{r}(t) = \frac{d}{dt} a(t) \vec{x}(t)$$

$$\dot{\vec{r}}(t) = \dot{a}(t) \vec{x}(t) + \vec{x}(t) \dot{a}(t)$$

$$\frac{\dot{\vec{r}}(t)}{a(t)} = \underbrace{\frac{\dot{a}(t)}{a(t)} \vec{x}(t)}_{\text{Hubble Flow term}} + \underbrace{\vec{x}(t) \dot{a}(t)}_{\text{peculiar velocity}}$$

This term is the Hubble Flow term. This represents the velocity due to expansion

This term is the peculiar velocity. This describes the local velocity of the object without considering the additional velocity due to expansion.

d. The name of this equation is the Zeldovich Approximation. This formula is written in co-moving coordinates.

$$\vec{x}(t) = \underbrace{\vec{q}}_{\text{initial position of Lagrangian coordinates}} + \underbrace{D_+(t)}_{\text{Growth rate of fluctuations}} \underbrace{\vec{\delta}_0(\vec{q})}_{\text{initial displacement}}$$

$$\dot{\vec{x}}(t) = \dot{\vec{q}} + \dot{D}_+(t) \vec{S}_0(\vec{q}) + D_+(t) \dot{\vec{S}}_0(\vec{q})$$

$$v_{pec}(t) = \dot{\vec{q}} + \dot{D}_+(t) \vec{S}_0(\vec{q}) + D_+(t) \dot{\vec{S}}_0(\vec{q})$$

e. We can measure the Hubble flow using redshift effects and peculiar velocities with radial velocities from spectroscopic data.

f. Gaussian Random Field is the name of the statistical distribution that describes the Universe's initial density field. The $P(k)$ function is

9

Part 2: General Relativity and Galaxies:

a.

$$g_{\mu\nu} = \begin{bmatrix} -(1 + \frac{2\Phi(\vec{r})}{c^2})c^2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} dt \\ dx \\ dy \\ dz \end{bmatrix}$$

This deviates from the simple Minkowski metric by including gravitational potential in the time coordinate, whereas the Minkowski metric looks like:

$$ds^2 = c^2 dt^2 + dx^2 + dy^2 + dz^2$$

$\Downarrow$

$$\gamma = \begin{bmatrix} c^2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} dt \\ dx \\ dy \\ dz \end{bmatrix}$$

This assumes a flat spacetime but in $g_{\mu\nu}$, it assumes a curved space

b.

$$\Phi(r) = \frac{-GM}{r}, \quad \int_{\mathbb{R}} \Phi(r)$$

$$G(r) = \frac{G \cdot M}{R^2}$$

$$= \phi(r) = - \int_{-\infty}^r g(r') dr'$$

c. Milky Way: $M \sim 10^{12} M_{\odot}$
 $R \sim 15 \text{ kpc}$

d. $-\frac{GM}{r} = \frac{-G(10^{12} M_{\odot})}{15 \text{ kpc}}$

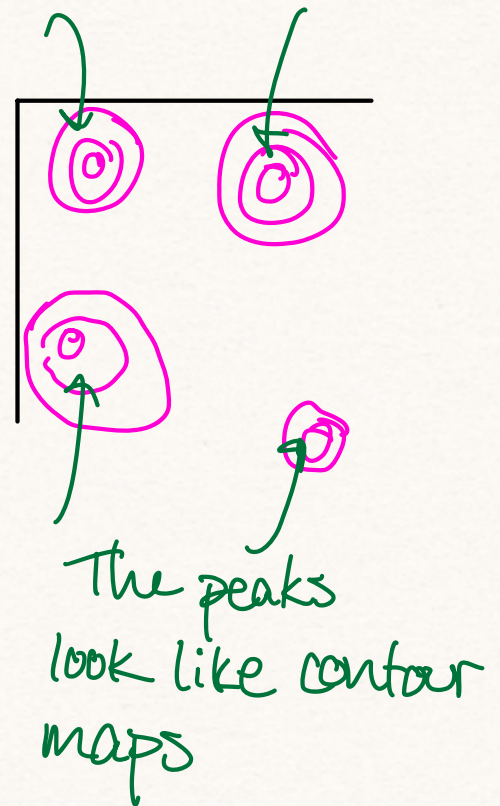
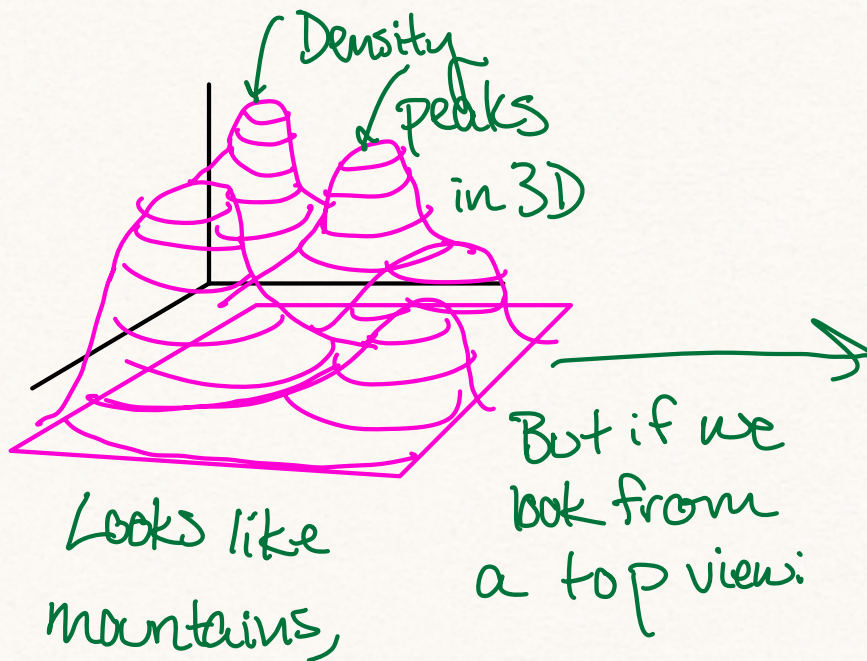
$$= \frac{(6.67 \times 10^{-11})(1.988 \times 10^{31})}{(4.628 \times 10^{20})}$$

$$\approx 2.8 \times 10^{22}$$

Part 3: Hydrodynamics of Galaxies

Exercise 2: Toomre Instability for Infinitely Thin Galactic Disk

a. Surface density is an area (so 2D dimensions) and takes into account the 3-D density (mass per unit area) and "flattens" it onto a plane.



$$\text{Surface Density} = \frac{\text{Mass}}{\text{length}^2}$$

b. Conservation of Mass:

$$\frac{\partial \Sigma_i}{\partial t} + \Sigma_0 (\partial_x v_{ix} + \partial_y v_{iy}) + v_{oy} \partial_y \Sigma_i = 0$$

Conservation of Momentum:

$$\frac{\partial v_{iy}}{\partial t} + \frac{k^2}{2\Omega_0} v_{ix} = - \sum_i \partial_y P_i - \partial_y \Phi_i$$

azimuthal
momentum

$$\frac{\partial v_{ix}}{\partial t} - 2\Omega_0 v_{iy} = - \sum_i \partial_x P_i - \partial_x \Phi_i$$

Radial
momentum

$$c. \frac{\partial \Sigma_i}{\partial t} + \sum_i (\partial_x v_{ix} + \cancel{\partial_y v_{iy}}) + \cancel{v_{iy} \partial_y \Sigma_i} = 0$$

$$\frac{\partial \Sigma_i}{\partial t} + \sum_i (\partial_x \underline{v_{ix}}) = 0$$

$$\frac{\partial \Sigma_i}{\partial t} + \sum_i (\partial_x \underline{v_{ix}^0 e^{i(\omega t - kx)}}) = 0$$

$$\frac{\partial \Sigma_i}{\partial t} - ik \sum_i v_{ix}^0 e^{i(\omega t - kx)} = 0$$

$$\cancel{\frac{\partial v_{iy}}{\partial t}} + \frac{k^2}{2\Omega_0} v_{ix} = - \cancel{\sum_i \partial_y P_i} - \cancel{\partial_y \Phi_i}$$

$$\frac{k^2}{2\Omega_0} v_{ix} = 0$$

$$\frac{k^2}{2\Omega_0} \left(v_{ix}^0 e^{i(\omega t - kx)} \right) = 0$$

$$\frac{\partial v_{ix}}{\partial t} - 2\Omega_0 v_{iy} = -\frac{1}{\Sigma_0} \partial_x P_1 - \partial_x \Phi_1$$

$$\frac{\partial}{\partial t} v_{ix} = -\frac{1}{\Sigma_0} \partial_x P_1 - \partial_x \Phi_1$$

$$\frac{\partial}{\partial t} \left(v_{ix}^0 e^{i(\omega t - kx)} \right) = -\frac{1}{\Sigma_0} \partial_x P_1 - \partial_x \Phi_1$$

$$i\omega v_{ix}^0 e^{i(\omega t - kx)} = -\frac{1}{\Sigma_0} \partial_x P_1 - \partial_x \Phi_1$$

$$\frac{\partial \Sigma_1}{\partial t} - ik \Sigma_0 v_{ix}^0 e^{i(\omega t - kx)} = 0$$

$$\frac{K^2}{2\Omega_0} \left(v_{ix}^0 e^{i(\omega t - kx)} \right) = 0$$

$$i\omega v_{ix}^0 e^{i(\omega t - kx)} = - \sum_0^1 \partial_x P_1 - \partial_x \Phi_1$$

$$\frac{d\Sigma_1}{dt} + \sum_0^1 (\partial_x v_{ix} + \cancel{\partial_y v_{iy}}) + v_{oy} \cancel{\partial_y \Sigma_1} = 0$$

$$\frac{\partial \Sigma_1}{\partial t} + \sum_0^1 \partial_x v_{ix} = 0 \quad \leftarrow \text{continuity}$$

$$\cancel{\frac{\partial v_{iy}}{\partial t}} + \frac{K^2}{2\Omega_0} v_{ix} = - \cancel{\sum_0^1 \partial_y P_1} - \cancel{\partial_y \Phi_1}$$

$$\cancel{\frac{\partial v_{iy}}{\partial t}} + \frac{K^2}{2\Omega_0} v_{ix} = 0 \quad \leftarrow \text{azimuthal momentum}$$

Radial

$$\frac{\partial v_{ix}}{\partial t} - 2\Omega_0 v_{iy} = -\frac{1}{\sum_0} \partial_x P_i - \partial_x \Phi_i$$

momentum

Continuity:

$$\frac{\partial}{\partial t} \left(\sum_i^0 e^{i(\omega t - kx)} \right) + \sum_0 \partial_x \left(v_{ix}^0 e^{i(\omega t - kx)} \right) = 0$$

$$i\omega \sum_i^0 e^{i(\omega t - kx)} - ik \sum_0 v_{ix}^0 e^{i(\omega t - kx)} = 0$$

$$\omega \sum_i^0 - k \sum_0 v_{ix}^0 = 0$$

$$v_{ix}^0 = \frac{\omega \sum_i^0}{k \sum_0}$$

Equation 1

$$\frac{\partial v_{iy}}{\partial t} + \frac{K^2}{2\Omega_0} v_{ix} = 0$$

$$\frac{\partial}{\partial t} \left(v_{iy}^0 e^{i(\omega t - kx)} \right) + \frac{K^2}{2\Omega_0} \left(v_{ix}^0 e^{i(\omega t - kx)} \right) = 0$$

$$i\omega v_{iy}^0 e^{i(\omega t - kx)} + \frac{k^2}{2\Omega_0} v_{ix}^0 e^{i(\omega t - kx)} = 0$$

$$i\omega v_{iy}^0 + \frac{k^2}{2\Omega_0} v_{ix}^0 = 0$$

$$v_{ix}^0 \frac{k^2}{2\Omega_0} = -i\omega v_{iy}^0$$

$$v_{iy}^0 = \frac{v_{ix}^0 k^2}{-2i\omega\Omega_0}$$

$$\frac{1}{i} \cdot \frac{i}{i} = \frac{i}{i^2}$$

$$\frac{i}{-1} = -i$$

$$v_{iy}^0 = \frac{i v_{ix}^0 k^2}{2\omega\Omega_0}$$

Equation 2

$$\frac{\partial v_{ix}}{\partial t} - 2\Omega_0 v_{iy} = -\frac{1}{\Sigma_0} \partial_x \mathcal{P}_1 - \partial_x \Phi_1$$

$$\frac{\partial}{\partial t} (v_{ix}^0 e^{i(\omega t - kx)}) - 2\Omega_0 (v_{iy}^0 e^{i(\omega t - kx)}) =$$

$$-\frac{1}{\Sigma_0} \partial_x (P_1^0 e^{i(\omega t - kx)}) - \partial_x (\Phi_1^0 e^{i(\omega t - kx)})$$

$$i\omega v_{1x}^0 e^{i(\omega t - kx)} - 2\Omega_0 v_{1y}^0 e^{i(\omega t - kx)} =$$

$$+ \frac{ik}{\Sigma_1} P_1^0 e^{i(\omega t - kx)} + ik \Phi_1^0 e^{i(\omega t - kx)}$$

$$i\omega v_{1x}^0 - 2\Omega_0 v_{1y}^0 = \frac{ikP_1^0}{\Sigma_1} + ik\Phi_1^0 \quad \text{Equation 3}$$

Plugging in Equation 2 into Equation 3:

$$i\omega v_{1x}^0 - 2\Omega_0 \left(\frac{i v_{1x}^0 H^2}{2\omega\Omega_0} \right) = \frac{ikP_1^0}{\Sigma_1} + ik\Phi_1^0$$

$$\omega v_{1x}^0 - \left(\frac{v_{1x}^0 H^2}{\omega} \right) = \frac{kP_1^0}{\Sigma_1} + k\Phi_1^0$$

$$\omega \left(\frac{\omega \Sigma_1^0}{k \Sigma_0} \right) - \left(\frac{\left(\frac{\omega \Sigma_1^0}{k \Sigma_0} \right) H^2}{\omega} \right) = \frac{kP_1^0}{\Sigma_1} + k\Phi_1^0$$

$$\frac{\Sigma_1^0}{K \Sigma_0^0} \omega^2 - \frac{\Sigma_1^0 H^2}{K \Sigma_0^0} = \frac{K P_1^0}{\Sigma_0^0} + K \underline{\Phi}_1^0$$

$$\frac{\Sigma_1^0}{\cancel{K \Sigma_0^0}} \omega^2 = \frac{\Sigma_1^0 H^2}{\cancel{K \Sigma_0^0}} + \frac{K}{\cancel{\Sigma_0^0}} P_1^0 + K \underline{\Phi}_1^0$$

$$\frac{\Sigma_1^0 \omega^2}{\Sigma_1^0} = \frac{\Sigma_1^0 H^2 + K^2 P_1^0 + K^2 \Sigma_0^0 \underline{\Phi}_1^0}{\cancel{\Sigma_1^0}}$$

$$\omega^2 = H^2 + \frac{K^2}{\Sigma_1^0} P_1^0 + \frac{K^2 \Sigma_0^0}{\Sigma_1^0} \underline{\Phi}_1^0$$

$$P_1^0 = C_S^2 \mathcal{P}_1^0$$

$$\omega^2 = H^2 + \frac{K^2}{\Sigma_1^0} (C_S^2 \mathcal{P}_1^0) + \frac{K^2 \Sigma_0^0}{\Sigma_1^0} \underline{\Phi}_1^0$$

We can conclude that $\mathcal{P}_1^0 = \Sigma_1^0$ because we are working with an infinitely thin disk, thus the 3D density $\mathcal{P}_1^0$ is really a 2D surface density.

$$\omega^2 = K^2 + c_s^2 K^2 + \frac{K^2 \Sigma'_0}{\Sigma'_1} \Phi_1^0$$

c_s in this case is the speed of sound, or the speed at which perturbations travel through the disk.

$$\omega^2 = K^2 + c_s^2 K^2 + \frac{K^2 \Sigma'_0}{\Sigma'_1} \left(- \frac{2\pi G \Sigma'_1}{|K|} \right)$$

$$\omega^2 = K^2 + c_s^2 K^2 - 2\pi G \Sigma'_0 |K|$$

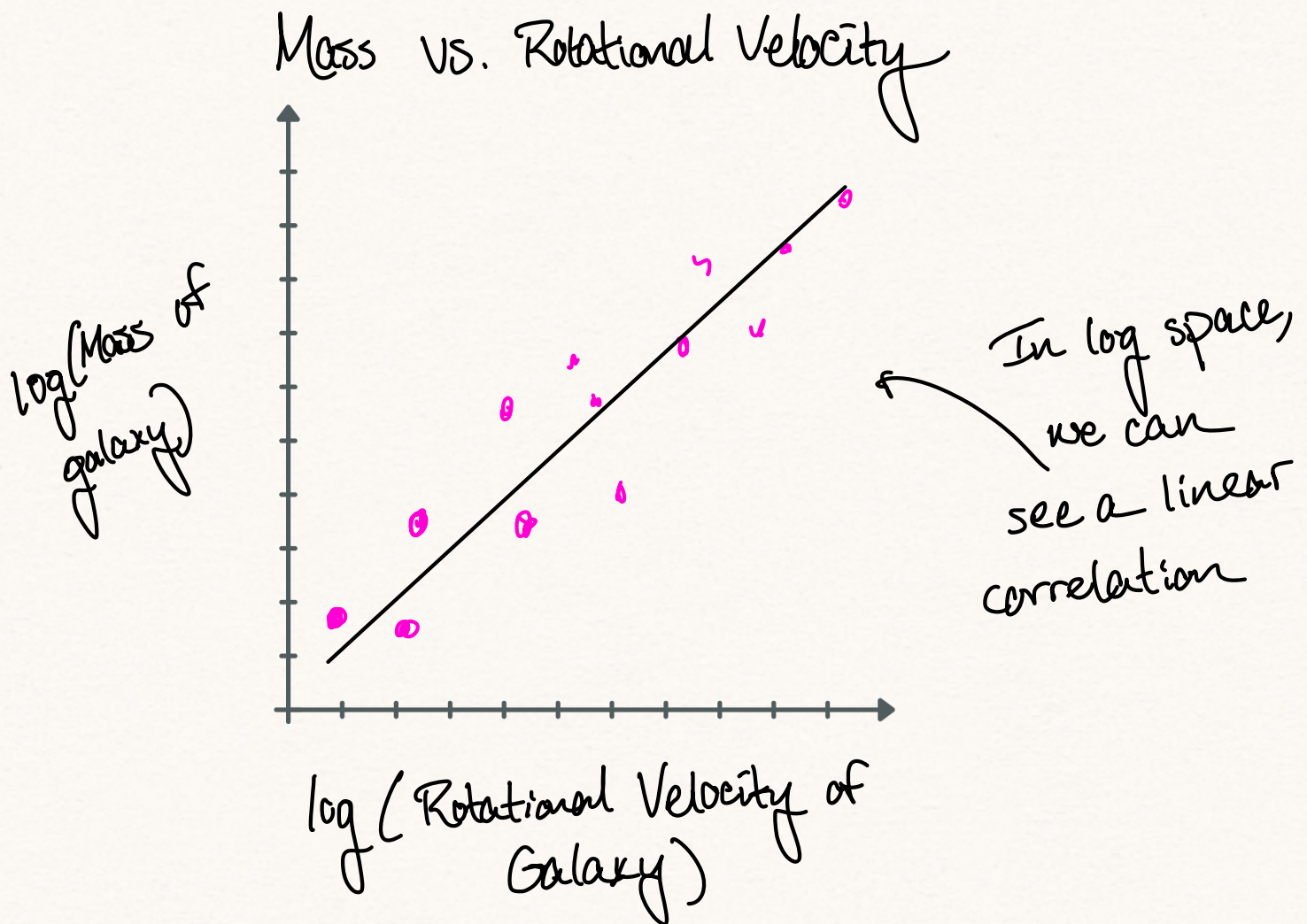
d. The Jeans Instability focuses on just the interaction between pressure and gravity while the Toomre Instability looks at rotational velocities, pressure, and gravity. The angular momentum is the "new force" (i.e. Coriolis

Effect) that competes against gravity

e. the new condition for the growing mode is when k^2 and the pressure term ($c_s^2 k^2$) is larger than the gravitational potential term. The threshold is the k^2 term which will dictate whether the disk remains stable ($\omega^2 \geq 1$) or unstable ($\omega^2 < 1$).

Part 4: Paper Analysis

a. The Tully-Fisher relation shows the relationship of luminosity vs. rotation rates in galaxies. This relationship does not include the baryonic gas and doesn't consider how that affects the relationship, whereas the Baryonic Tully-Fisher relationship does. In general, the mass of the galaxy can also affect rotational velocities.



b. The study tries to understand galaxy evolution through the lens of using the BTFR, since this relationship arises from the underlying physics of galaxy evolution. Because the BTFR is used often to constrain and test galaxy formation and evolution, this paper wants to study the intrinsic scatter of the BTFR because it can lead to imposing strong constraints on galaxy formation and evolution models. This paper wants to see how the intrinsic scatter, and also the slope, of the relationship varies if different sample sizes were used to measure the BTFR.

C. The Eagles simulation data was used in this paper. The strengths of this dataset are:

1. Large amount of datapoints (aka: galaxies) that is not available with observational data.
2. Can choose varying resolutions of the dataset, depending what you want to explore.

The downside of this dataset:

1. Because this is a simulated dataset, the data may not be as true to observational data.
2. This paper chooses to use all of the Eagles simulated data, which can lower the resolution, which may make the data less

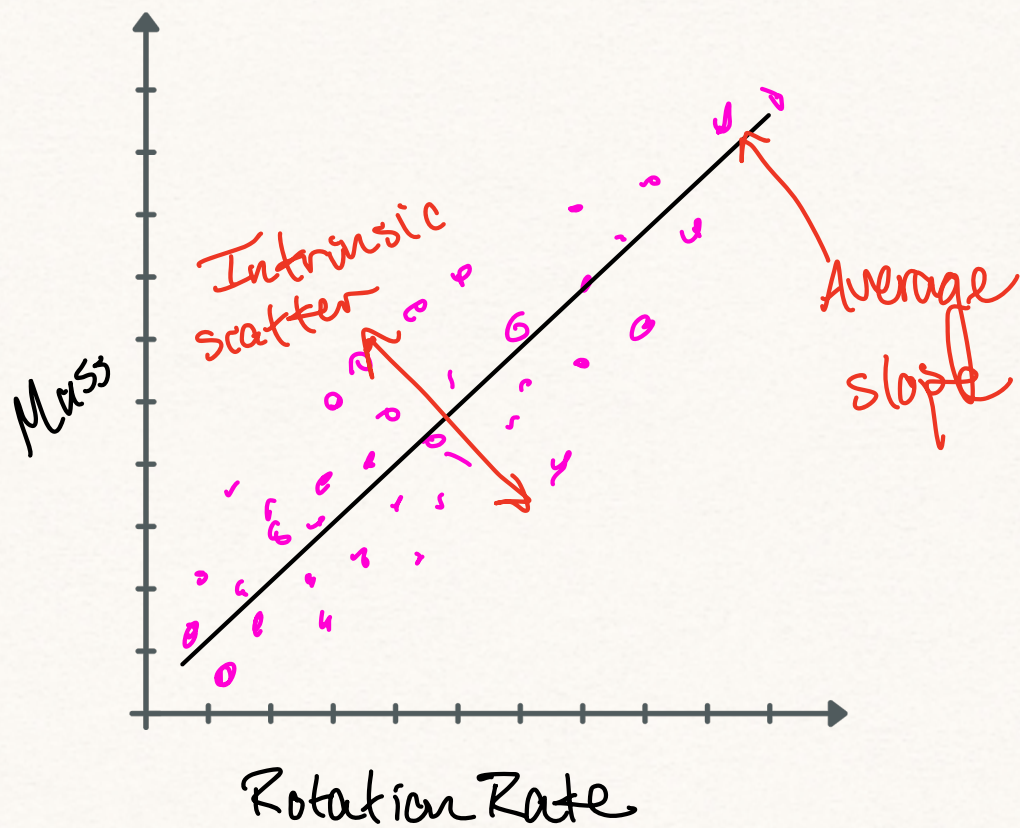
accurate.

d. $v_{\max}$ is the maximum value of the circular velocity of the rotation rate of the particles in the galaxy. It is obtained by taking the total enclosed mass in a sphere with radius r .

$$v_{\max} = G[M(<r)/r]^{1/2}.$$

this is different from observation since observational calculations can use spectral data to determine the rotation rate.

e. The key parameters of the BTFR are the slope and the intrinsic scatter. These parameters are affected by rotational rates, masses, and even the sample size of the data used to calculate the BTFR.



These parameters are important to fully understand and refine since scientists rely on this relationship in simulations to - then better understand galaxy formation and evolution.

F. These key parameters change with your sample size because by adding in more data, you will have a larger range of numbers to test the BTFR. Smaller sample sizes lead to larger standard deviations of the BTFR, while larger sample sizes do the opposite. The smaller the data set you use, the less diverse your galaxies will be, causing the standard deviation of the intrinsic scatter to increase because of the biased data.

Slopes can become higher or lower depending on the sample size too. It seems from this paper, calculations of the slope with observational data (which is smaller than the simulated data) tends to be higher

than when calculated with simulated data.

g. The smaller the value of $v_{\max}$, the larger the standard deviation of the intrinsic scatter and the larger the slope. This is, again, due to the bias in the dataset if you only include low rotation rate galaxies.

h. The main conclusions of this paper are sample size of galaxies are a new parameter that should be added to the list of consideration when measuring the BTFR. The sample size should be large enough to include a diverse population of galaxy parameters to prevent biasing your

results.

i. The Λ CDM references the current cosmological model of the Universe. It specifically includes dark energy and cold dark matter and how those things influence what we see in the Universe today. The BTFR, specifically its intrinsic scatter can be used to test the Λ CDM model because currently, observational data used to derive the scatter does not match the Λ CDM model predictions, however because the paper concludes that sample size affects the scatter, this may be the cause for why observational data does

not currently match the LCDll model.